

Coding & Solution

Date	12 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/24A31A0597/OptiCrop-AI-Powered-Crop-Recommendation-System.git
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Crop recommendation, Input validation, ML prediction, Confidence score, Responsive UI
Pending / Incomplete Features	User login, Database integration, Weather API integration
Setup / Run Instructions	1. Install dependencies (pip install -r requirements.txt) 2. Run python app.py 3. Open the local or deployed URL.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The solution follows clean coding practices, modular architecture, and version control using GitHub.
